

Describing Data Sources

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Motivation and Outline

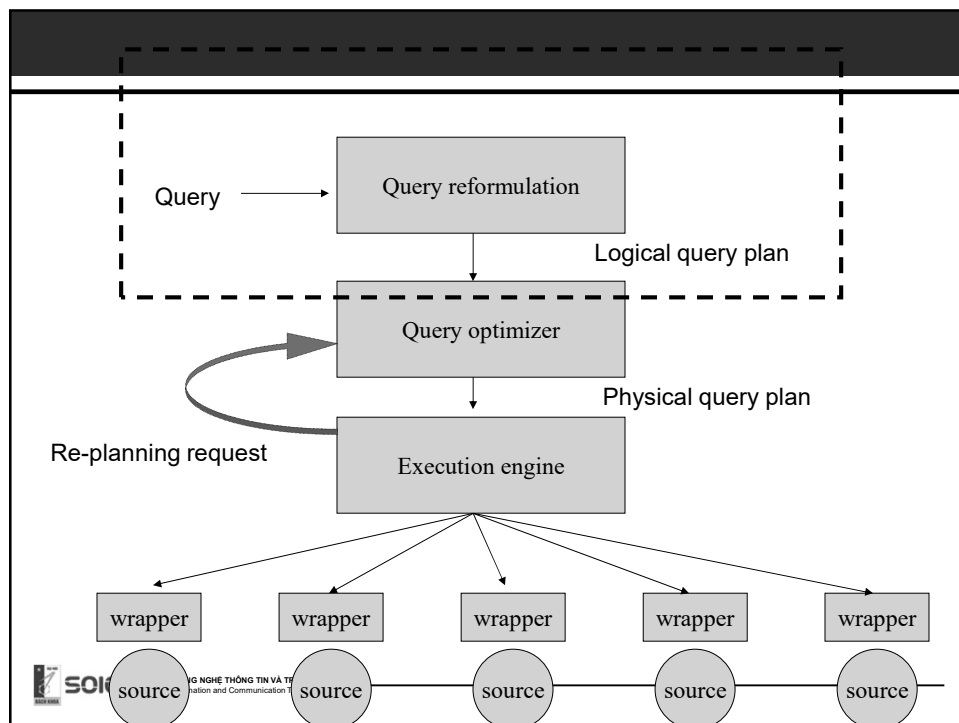
- Descriptions of data sources enable the data integration system to:
 - Determine which sources are relevant to a query
 - Access the sources appropriately
 - Combine data from multiple sources
 - Overcome limitations that specific sources may have
 - Identify opportunities for more efficient query evaluation
- Source descriptions are a formalism for specifying the important aspects of data sources.

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Outline

- Introduction to semantic heterogeneity
 - Schema mapping languages
 - Query unfolding

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Schema Heterogeneity

- Schema heterogeneity is a fact of life.
 - Whenever schemas are designed by different people/organizations, they will be different, even if they model the *same* domain!
- The goal of schema mappings is to reconcile schema heterogeneity:
 - Mostly between the mediated schema and the schema of the data sources.

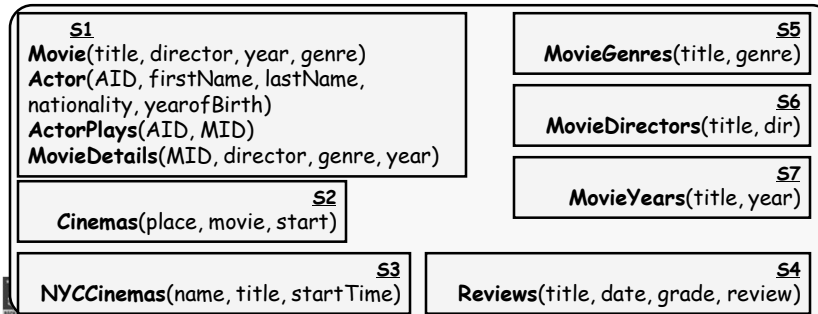
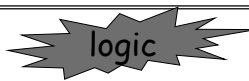
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Schema Heterogeneity by Example

Mediated Schema

Movie: title, director, year, genre
 Actors: title, name
 Plays: movie, location, startTime
 Reviews: title, rating, description

Sources



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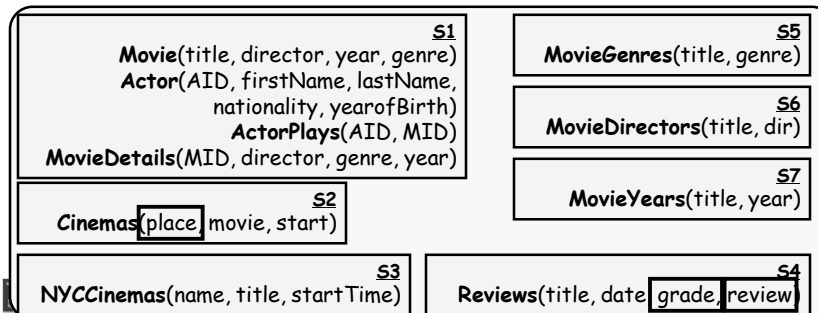
Table and Attribute Naming

Mediated Schema

Movie: title, director, year, genre
 Actors: title, name
 Plays: movie, location, startTime
 Reviews: title, rating, description

Sources

Table and attribute names



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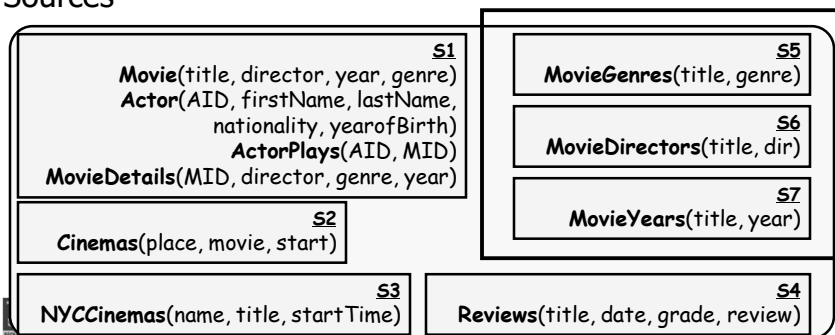
Tabular Organization of Schema

Mediated Schema

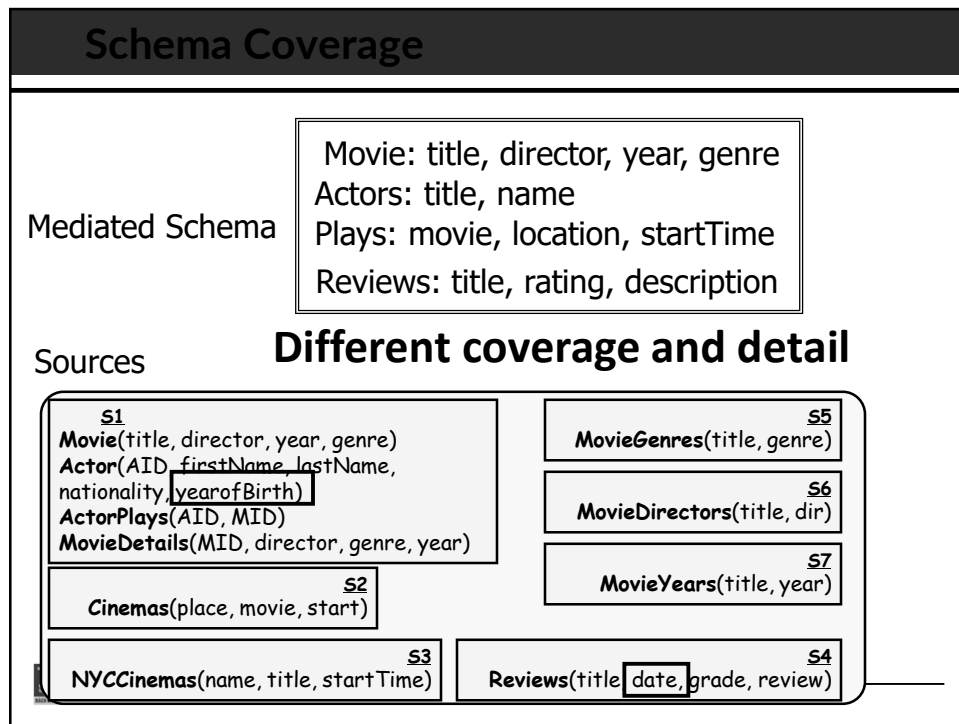
Movie: title, director, year, genre
 Actors: title, name
 Plays: movie, location, startTime
 Reviews: title, rating, description

Sources

Different tabular organization



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Semantic Heterogeneity Summary

- Differences in:
 - Naming of schema elements
 - Organization of tables
 - Coverage and detail of schema
 - Data-level representation (IBM vs. International Business Machines)
- Reason:
 - Schemas probably designed for different applications/contexts.

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More in Source Descriptions

- Possible access patterns to the data source:
 - Are there required inputs? (e.g., web forms, web services)
 - Can the source process complex database queries?
- Source completeness:
 - Is the source complete, or partially complete?
- Reliability, load restrictions, mirror sites, ...

Outline

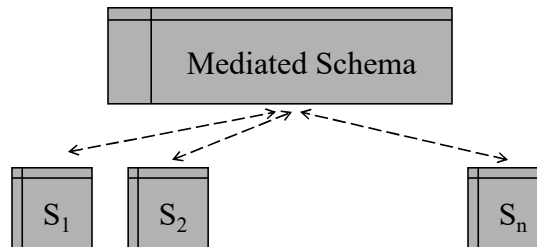
- ✓ Introduction to semantic heterogeneity
- Schema mapping languages
- Query unfolding

Principles of Schema Mappings

Schema mappings describe the relationship between:



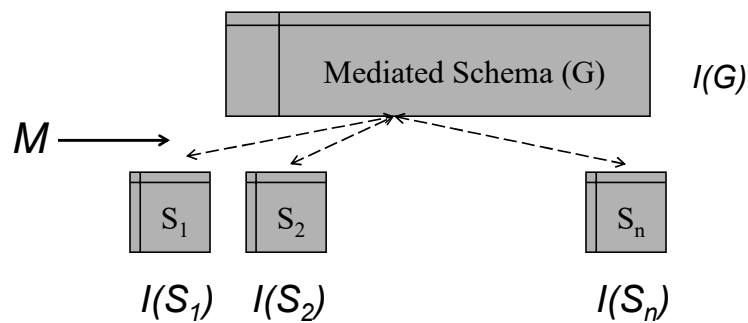
Or between:



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Semantics of Schema Mappings

Formally, schema mappings describe a relation: which instances of the mediated schema are consistent with the current instances of the data sources.



$I(G)$ ($I(S_i)$): the set of possible instances of the schema G (S_i).

$$M_R \subseteq I(G) \times I(S_1) \times \dots \times I(S_n)$$

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Relations, explained

- A relation is a subset of the Cartesian product of its columns' domains:

1	1
2	4
3	9
4	16
5	25

The table on the left is a subset of the Cartesian product of

Int x Int.

The table describes the Squared relation.

Similarly, M_r specifies the possible instances of the mediated schema, given instances of the sources.

Possible Instances of Mediated Schema: Simple Example

- Source 1: (Director, Title, Year) with tuples
 - {(Allen, Manhattan, 1979),
 - (Coppola, GodFather, 1972)}
- Mediated schema: (Title, Year)
 - Simple projection of Source 1
 - {(Manhattan, 1979), (GodFather, 1972)}

Possible Instances of Mediated Schema: Second Example

- Source 1: (Title, Year) with tuples
 - {(Manhattan, 1979), (GodFather, 1972)}
- Mediated schema: (Director, Title, Year)
 - Possible instance 1: {(Allen, Manhattan, 1979), (Coppola, GodFather, 1972)}
 - Another possible instance 2: {(Halevy, Manhattan, 1979), (Stonebraker, GodFather, 1972)}.
- This matters when we answer queries:
 - See next slide.



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Answering Queries over Possible Instances of Mediated Schema

- Mediated schema: (Director, Title, Year)
 - Possible instance 1: {(Allen, Manhattan, 1979), (Coppola, GodFather, 1972)}
 - Another possible instance 2: {(Halevy, Manhattan, 1979), (Stonebraker, GodFather, 1972)}.
- Query Q1: return all years of movies
 - Answer: (1979, 1972) are certain answers.
- Query Q2: return all directors
 - No certain answers because no directors appear in all possible instances of the mediated schema.

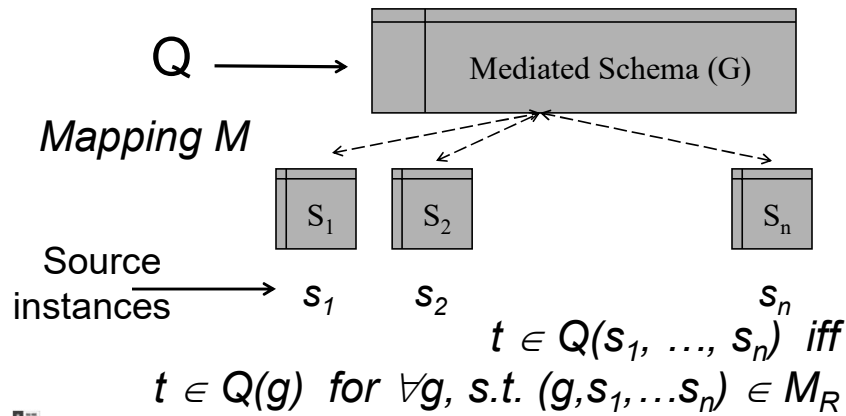


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Certain Answers Makes this Formal

An answer is certain if it is true in every instance of the mediated schema that is consistent with: (1) the instances of the sources and (2) the mapping M .



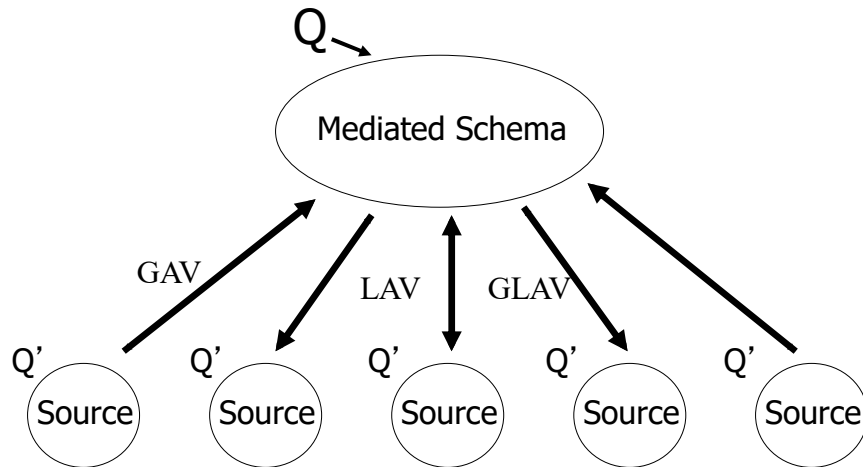
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Desiderata from Source Description Languages

- Flexibility:
 - Should be able to express relationships between real schemata
- Efficient reformulation:
 - Computational complexity of reformulation and finding answers
- Easy update:
 - Should be easy to add and delete sources

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Languages for Schema Mapping



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Global-as-View (GAV)

- Mediated schema defined as a set of views over the data sources

Movie: title, director, year, genre

S1
Movie(MID,title)
Actor(AID, firstName, lastName,
 nationality, yearofBirth)
ActorPlays(AID, MID)
MovieDetails(MID, director, genre, year)

Movie(title,director,year,genre) $\supseteq$

S1.Movie(MID,title),

S1.MovieDetail(MID,director,genre,year)

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GAV: Formal Definition

A set of expressions of the form:

$$G_i(\bar{X}) \supseteq Q(\bar{S}) \quad \text{or} \quad G_i(\bar{X}) = Q(\bar{S})$$

open-world
assumption

closed-world
assumption

- G_i : relation in mediated schema
- $Q(\bar{S})$: query over source relations

GAV Example

Movie: title, director, year, genre

$Movie(title, director, year, genre) \supseteq$
 $S1.Movie(MID, title), S1.MovieDetail(MID, director, genre, year)$

$Movie(title, director, year, genre) \supseteq$
 $S5.MovieGenres(title, genre),$
 $S6.MovieDirectors(title, director),$
 $S7.MovieYears(title, year)$

GAV Example (cont.)

Plays: movie, location, startTime

$Plays(movie, location, startTime) \supseteq$
 $S2.Cinemas(location, movie, startTime)$

$Plays(movie, location, startTime) \supseteq$
 $S3.NYCCinemas(location, movie, startTime)$

S2
Cinemas(place, movie, start)

S3
NYCCinemas(name, title, startTime)

Reformulation in GAV

- Given a query Q on the mediated schema:
 - Return the best query possible on the data sources.

Reformulation in GAV= Query/View Unfolding

Movie: title, director, year, genre

Plays: movie, location, startTime

$Q(\text{title}, \text{location}, \text{startTime}) :-$

$\text{Movie}(\text{title}, \text{director}, \text{year}, \text{"comedy"}),$

$\text{Plays}(\text{title}, \text{location}, \text{st}), \text{st} \geq 8\text{pm}$

$\text{Movie}(\text{title}, \text{director}, \text{year}, \text{genre}) \supseteq$

$S1.\text{Movie}(\text{MID}, \text{title}), S1.\text{MovieDetail}(\text{MID}, \text{director}, \text{genre}, \text{year})$

$\text{Plays}(\text{movie}, \text{location}, \text{startTime}) \supseteq$

$S2.\text{Cinemas}(\text{location}, \text{movie}, \text{startTime})$



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First Reformulation

$Q(\text{title}, \text{location}, \text{startTime}) :-$

$\text{Movie}(\text{title}, \text{director}, \text{year}, \text{"comedy"}),$

$\text{Plays}(\text{title}, \text{location}, \text{st}), \text{st} \geq 8\text{pm}$



$Q'(\text{title}, \text{location}, \text{startTime}) :-$

$S1.\text{Movie}(\text{MID}, \text{title}),$

$S1.\text{MovieDetail}(\text{MID}, \text{director}, \text{"comedy"}, \text{year})$

$S2.\text{Cinemas}(\text{location}, \text{title}, \text{st}), \text{st} \geq 8\text{pm}$



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Another Reformulation

$Q(\text{title}, \text{location}, \text{startTime}) :-$

$\text{Movie}(\text{title}, \text{director}, \text{year}, \text{"comedy"}),$

$\text{Plays}(\text{title}, \text{location}, \text{st}), \text{st} \geq 8\text{pm}$



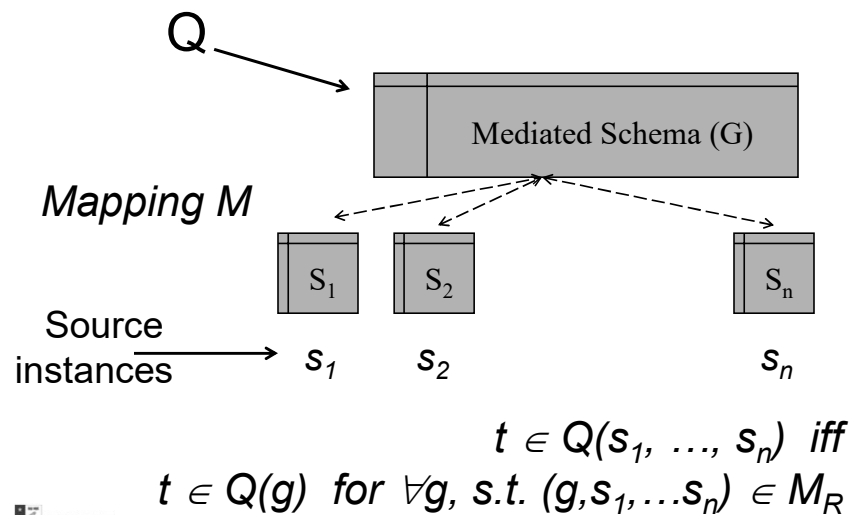
$Q'(\text{title}, \text{location}, \text{startTime}) :-$

$S1.\text{Movie}(\text{MID}, \text{title}),$

$S1.\text{MovieDetail}(\text{MID}, \text{director}, \text{"comedy"}, \text{year})$

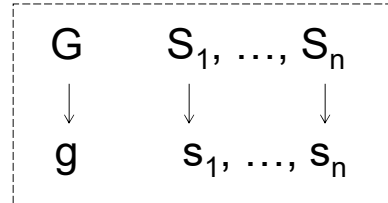
$S3.\text{NYCCinemas}(\text{location}, \text{title}, \text{st}), \text{st} \geq 8\text{pm}$

Certain Answers (recall definition)



Semantics of GAV

$(g, s_1, \dots, s_n) \in M_R$ if:



$$G_i(\bar{X}) \supseteq Q(\bar{S}) \Rightarrow$$

The extension of G_i in g is a super-set of evaluating Q_i on the sources.

$$G_i(\bar{X}) = Q(\bar{S}) \Rightarrow$$

The extension of G_i in g is equal to evaluating Q_i on the sources.

Tricky Example for GAV

S8: stores pairs of (actor, director)

Movie: title, director, year, genre
Actors: title, name
Plays: movie, location, startTime
Reviews: title, rating, description

$$Actors(NULL, actor) \supseteq S8(actor, director)$$

$$Movie(NULL, director, NULL, NULL) \supseteq S8(actor, director)$$

Tricky Example for GAV

$Actors(NULL, actor) \supseteq S8(actor, director)$

$Movie(NULL, director, NULL, NULL) \supseteq S8(actor, director)$

Given the S8 tuples:

$(\{Keaton, Allen\}, \{Pacino, Coppola\})$

We'd get tuples for the mediated schema:

Actors: $(\{NULL, Keaton\}, \{NULL, Pacino\})$

Movie: $(\{NULL, Allen, NULL, NULL\},$
 $\{NULL, Coppola, NULL, NULL\})$

Tricky Example (2)

Actors: $(\{NULL, Keaton\}, \{NULL, Pacino\})$

Movie: $(\{NULL, Allen, NULL, NULL\},$
 $\{NULL, Coppola, NULL, NULL\})$

Can't answer the query:

Q(actor, director) :-

Actors(title, actor),

Movie(title, director, genre, year)

LAV (Local as View) will solve this problem

GAV Summary

- Mediated schema is defined as views over the sources.
- Reformulation is conceptually easy
 - Polynomial-time reformulation and query answering.
- GAV forces everything into the mediated schema's perspective:
 - Cannot capture a variety of tabular organizations.



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Local-as-View (LAV)

- Data sources defined as views over mediated schema!

S5
MovieGenres(title, genre)

S6
MovieDirectors(title, dir)

S7
MovieYears(title, year)

Movie: title, director, year, genre
Actors: title, name
Plays: movie, location, startTime
Reviews: title, rating, description

$S5.MovieGenres(title, genre) \subseteq Movie(title, dir, year, genre)$

$S6.MovieDirectors(title, dir) \subseteq Movie(title, dir, year, genre)$



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Local-as-View (LAV)

- Data sources defined as views over mediated schema!

S8
ActorDirectors(actor,dir)

Movie: title, director, year, genre
Actors: title, name
Plays: movie, location, startTime
Reviews: title, rating, description

$S8.ActorDirectors(actor,dir) \subseteq$
 $Movie(title,dir,year,genre), Actor(title,actor)$
 $year \geq 1980$

LAV: Formal Definition

A set of expressions of the form:

$$S_i(\bar{X}) \subseteq Q_i(G) \text{ or } S_i(\bar{X}) = Q_i(G)$$

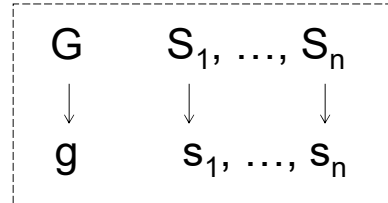
open-world
assumption

closed-world
assumption

- $S_i(\bar{X})$: source relation
- $Q_i(G)$: query over mediated schema

Semantics of LAV

$(g, s_1, \dots, s_n) \in M_R$ if:



$S_i(\bar{X}) \subseteq Q_i(G) \Rightarrow$
The result of Q_i over g is a superset of s_i .

$S_i(\bar{X}) = Q_i(G) \Rightarrow$
The result of Q_i over g equals s_i .

Possible Databases

Unlike GAV, LAV definitions imply a *set* of possible databases for the mediated schema.

$S8.ActorDirectors(actor, dir) \subseteq$

$Movie(title, dir, year, genre), Actor(title, actor)$

$S8 : \{(Keaton, Allen)\}$

Two possible databases for the mediate schema are:

Movie: {("manhattan", allen, 1979, comedy)}

Actor: {("manhattan", keaton)}

Movie: {("foobar", allen, 1979, comedy)}

Actor: {("foobar", keaton)}

Possible Databases

Since the source may be incomplete, other tuples may be in the instance of the mediated schema:

$S8.ActorDirectors(actor, dir) \subseteq$

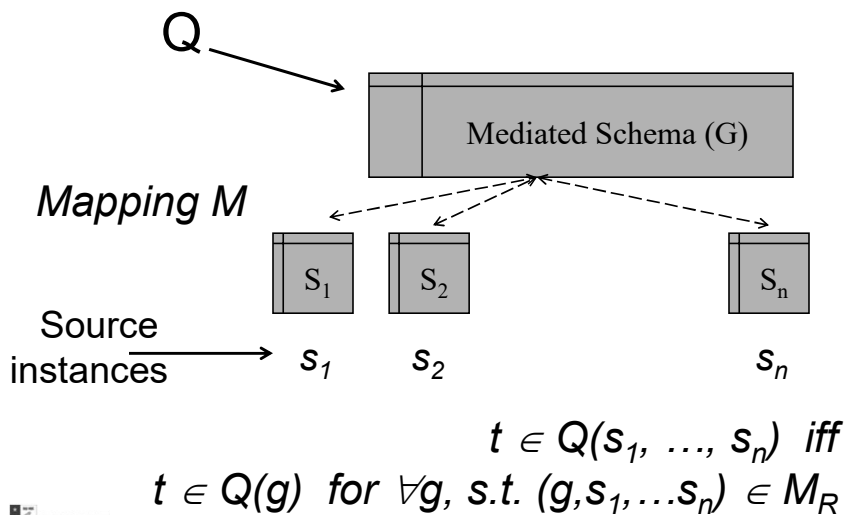
$Movie(title, dir, year, genre), Actor(title, actor)$

$S8 : \{(Keaton, Allen)\}$

Movie: {(manhattan, allen, 1981, comedy),
(leatherheads, clooney, 2008, comedy)}

Actor: {(manhattan, keaton),
(the godfather, keaton)}

Certain Answers: by now you know this slide by heart



Certain Answers Example

$S8.ActorDirectors(actor, dir) \subseteq$
 $Movie(title, dir, year, genre), Actor(title, actor)$
 $year \geq 1980$

$S8 : \{(Keaton, Allen)\}$

$Q(actor, dir) : -$

$Movie(title, dir, year, genre), Actor(title, actor)$

Only one certain answer: $(Keaton, Allen)$



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Certain Answers Example 2

$V_8(dir) = DirectorActor(ID, dir, actor) \quad \{Allen\}$

$V_9(actor) = DirectorActor(ID, dir, actor) \quad \{Keaton\}$

$Q(dir, actor) : -Director(ID, dir), Actor(ID, actor)$

Under closed-world assumption:
single DB possible $\Rightarrow (Allen, Keaton)$

Under open-world assumption:
no certain answers.

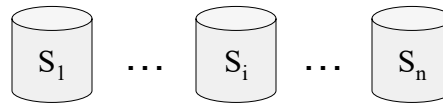


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Reformulation in LAV

We're given tuples for
sources
(expressed as views)



Mediated schema
(but no tuples)

Query over mediated schema

This is exactly the problem of:...
Answering queries using views!

Reformulation in LAV

Q(title) :-

Movie(title, director, year, "comedy"), year >=1960

Q'(title) :-

S5.MovieGenres(title, "comedy"),
S7.MovieYears(title, year), year >=1960

Q'(title) :-

S9(title, year, "comedy")

Local-as-View Summary

- Reformulation = answering queries using views
- Algorithms work well in practice:
 - Reformulation is not the bottleneck
- Under some conditions, guaranteed to find all certain answers
 - In practice, they typically do.
- LAV expresses incomplete information
 - GAV does not. Only a single instance of the mediated schema is consistent with sources.



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LAV Limitation

Movie: title, director, year, genre

S1

Movie(MID,title)

Actor(AID, firstName, lastName,nationality,yearofBirth)

ActorPlays(AID, MID)

MovieDetails(MID, director, genre, year)

If a key is internal to a data source, LAV cannot use it.



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GLAV[The Best of all Worlds*]

A set of expressions of the form:

$$Q^S(\bar{X}) \subseteq Q^G(\bar{X}) \text{ or } Q^S(\bar{X}) = Q^G(\bar{X})$$

- Q^G : query over mediated schema
- Q^S : query over data sources

*Apologies to the acronym Gods...

GLAV Example

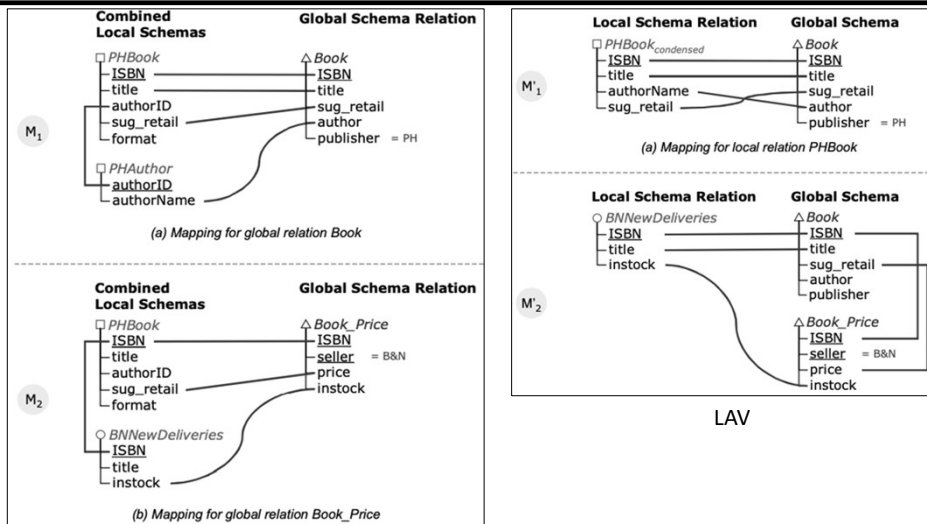
$$\begin{aligned} &S1.Movie(MID, title), S1.MovieDetail(MID, dir, genre, year) \\ &\subseteq \\ &Movie(title, dir, "comedy", year), year \geq 1970 \end{aligned}$$

Reformulation in GLAV

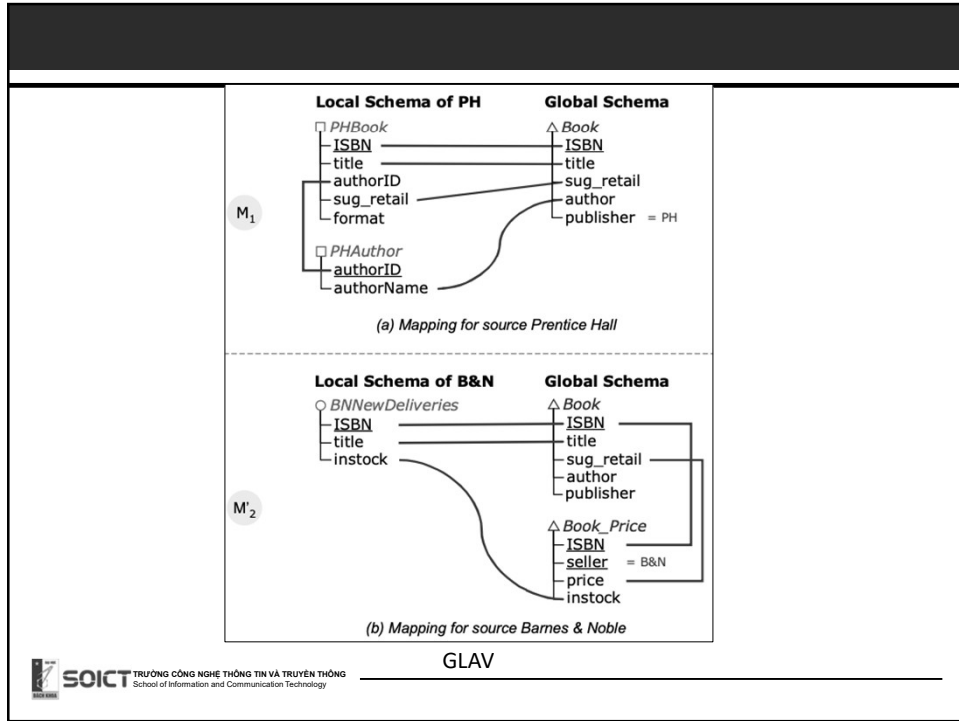
- Given a query Q
- Find a rewriting Q' using the views

$$Q_1^G, \dots, Q_n^G \quad \boxed{Q^S(\bar{X}) \subseteq Q^G(\bar{X})}$$

- Create Q'' by replacing: $Q_i^G \rightarrow Q_i^S$
- Unfold $Q_1^S, \dots, Q_n^S$



GAV



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An Alternative Notation for GLAV: Tuple Generating Dependencies

Tuple generating dependencies (Chapter 2.1.2) can be used to specify GLAV expressions.

The TGD

$$(\forall \bar{X}) s_1(\bar{X}_1), \dots, s_m(\bar{X}_m) \rightarrow (\exists \bar{Y}) t_1(\bar{Y}_1), \dots, t_k(\bar{Y}_k)$$

is equivalent to the GLAV expression: $Q^S(\bar{X}) \subseteq Q^G(\bar{X})$

where: $Q^S(\bar{X}) : -s_1(\bar{X}_1), \dots, s_m(\bar{X}_m)$

$Q^G(\bar{Y}) : -t_1(\bar{Y}_1), \dots, t_k(\bar{Y}_k)$

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GLAV -> TGD Example

$$S1.Movie(MID, title), S1.MovieDetail(MID, dir, genre, year) \\ \subseteq \\ Movie(title, dir, "comedy", year), year \geq 1970$$

→

$$S1.Movie(MID, title) \wedge S1.MovieDetail(MID, dir, genre, year) \\ \rightarrow Movie(title, dir, "comedy", year) \wedge year \geq 1970$$

Reformulation with TGD descriptions can be done relatively directly with the Inverse Rules Algorithm

Outline

- ✓ Introduction to semantic heterogeneity
- ✓ Schema mapping languages
- Query unfolding

Query Unfolding

- Query composition is an important mechanism for writing complex queries.
 - Build query from views in a bottom up fashion.
- Query unfolding “unwinds” query composition.
- Important for:
 - Comparing between queries expressed with views
 - Query optimization (to examine all possible join orders)
 - Unfolding may even discover that the composition of two satisfiable queries is unsatisfiable! (exercise: find such an example).

Query Unfolding Example

$Q_1(X, Y) : \neg \text{Flight}(X, Z), \text{Hub}(Z), \text{Flight}(Z, Y)$

$Q_2(X, Y) : \neg \text{Hub}(Z), \text{Flight}(Z, X), \text{Flight}(X, Y)$

$Q_3(X, Z) : \neg Q_1(X, Y), Q_2(Y, Z)$

The unfolding of Q_3 is:

$Q'_3(X, Z) : \neg \text{Flight}(X, U), \text{Hub}(U), \text{Flight}(U, Y),$
 $\text{Hub}(W), \text{Flight}(W, Y), \text{Flight}(X, Z)$

Query Unfolding Algorithm

- Find a subgoal $p(X_1, \dots, X_n)$ such that p is defined by a rule r .
- Unify $p(X_1, \dots, X_n)$ with the head of r .
- Replace $p(X_1, \dots, X_n)$ with the result of applying the unifier to the subgoals of r (use fresh variables for the existential variables of r).
- Iterate until no unifications can be found.
- If p is defined by a union of $r_1, \dots, r_n$, create n rules, for each of the r 's.

Query Unfolding Summary

- Unfolding does not necessarily create a more efficient query!
 - Just lets the optimizer explore more evaluation strategies.
 - Unfolding is the opposite of rewriting queries using views
- The size of the resulting query can grow exponentially